power

File: power.kicad_sch

mcu

File: mcu.kicad_sch

sensing

File: sensing.kicad_sch

vio

File: vio.kicad_sch

translators

File: translators.kicad_sch

connectors

File: connectors.kicad_sch

range_5v_sw

File: range_5v_sw.kicad_sch

range_3v3_sw

File: range_3v3_sw.kicad_sch

readback

File: readback.kicad_sch

OOTY rev A — hierarchical schematic generated by scripts/build_schematic.py

Specification: SPEC.md — design notes and calculations: NOTES.md

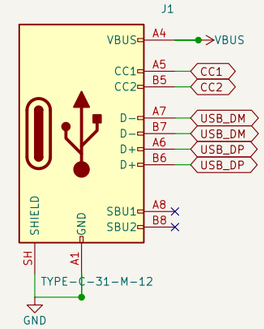
OOTY

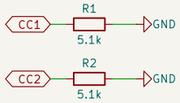
Sheet: /
File: OOTY.kicad_sch

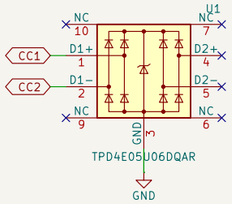
Title: OOTY — USB Hardware-in-the-Loop board

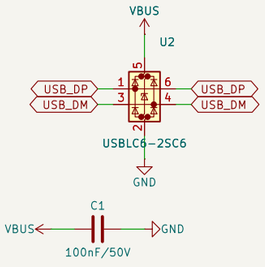
Size: A4	Date: 2026-10-20	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 1/10

USB-C input

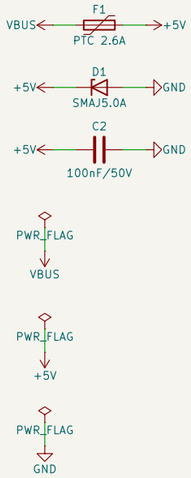








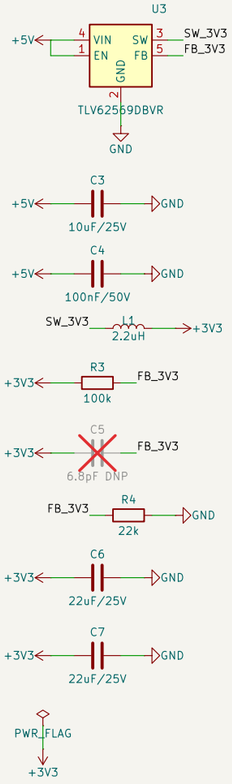
VBUS protection



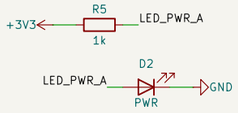
Rail test points



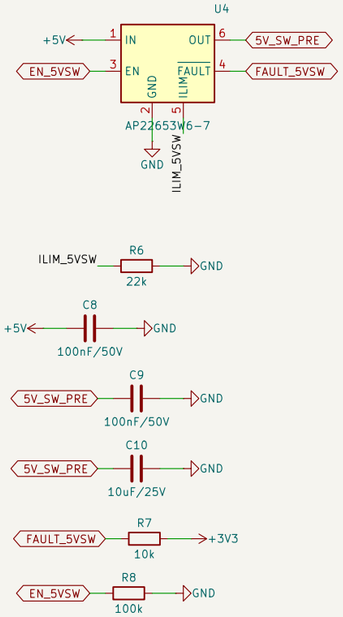
3V3 buck (TLV62569)



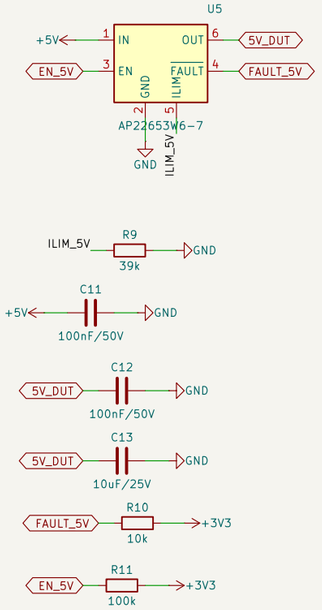
Power OK LED



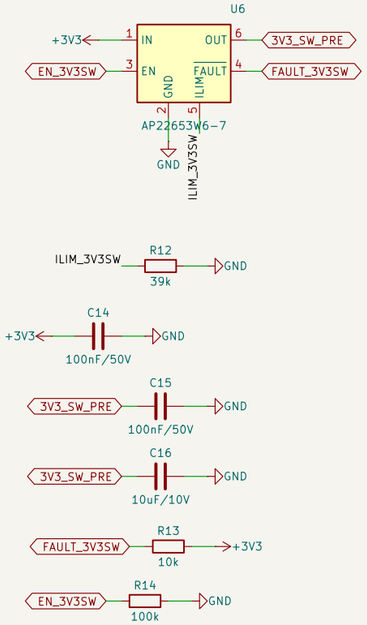
Switch 5V_SW (switched, 1 A)



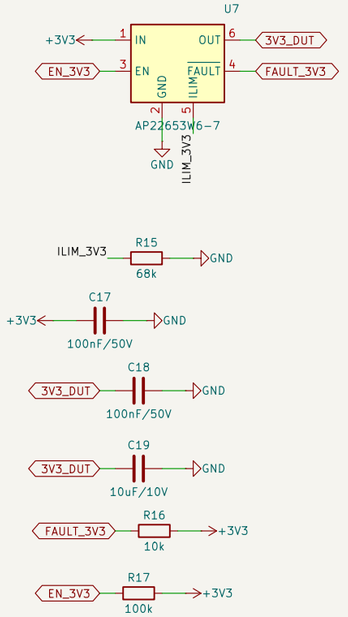
Switch 5V fixed (0.5 A)



Switch 3V3_SW (switched, 0.5 A)



Switch 3V3 fixed (0.3 A)

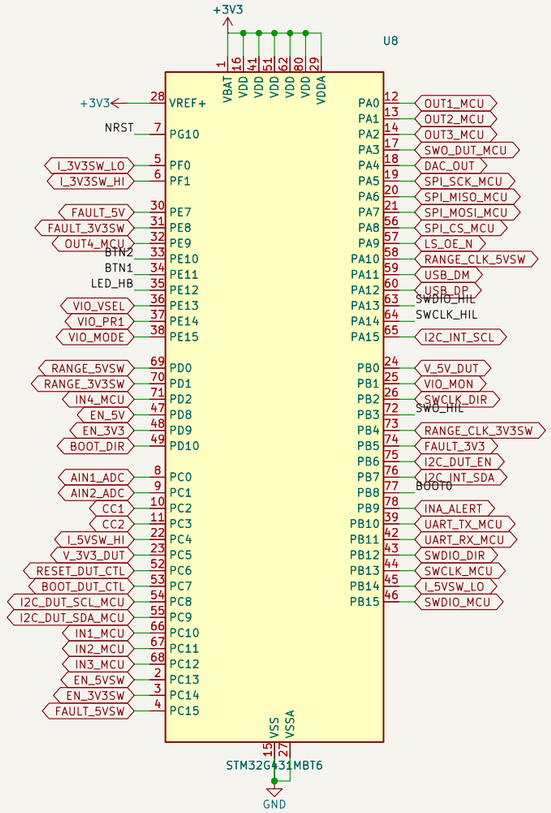


AP22653 ILIM (DS41186 p.10, VIN = 5 V): 22k -> 1.16 A typ, 39k -> 0.64 A typ, 68k -> 0.35 A typ. See NOTES.md.

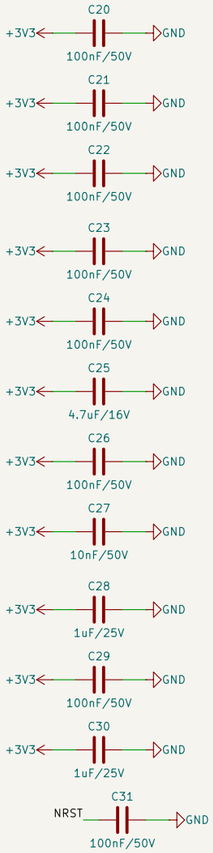
Buck: Vout = 0.6 V x (1 + 100k/22k) = 3.327 V.

OOTY		
Sheet: /power/		
File: power.kicad_sch		
Title: Power: USB-C, 3V3 buck, DUT rails		
Size: A3	Date: 2026-10-20	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 2/10

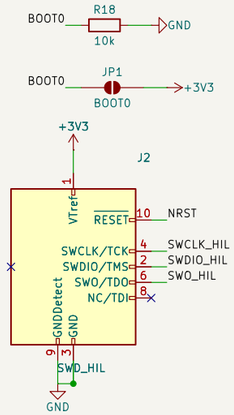
STM32G431MBT6



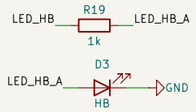
Decoupling (DS12589 5.1.6)



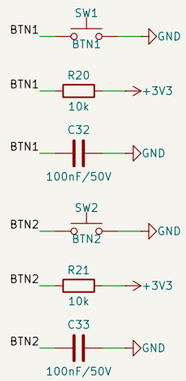
Boot / HIL debug (SWD)



Heartbeat LED



User buttons



USB FS runs crystal-less (HSI48 + CRS). BOOT0 solder jumper: bridge it to force the system bootloader.

PB6 may carry a UCPD dead-battery pull-down at reset (PA9 = DBCC1 is pulled high); I2C_DUT_EN has an external pull-down anyway. Same for PB4 (CC2): PA10 = DBCC2 is held low by the 100k on RANGE_CLK_5VSW; PB4 also has the NJTRST pull-up at reset. Firmware sets PWR_CR3.UCPD1_DBDIS first, then drives PB4 low.

Pin choice follows the PCB floor plan (MCU rotated 180 deg): see NOTES.md.

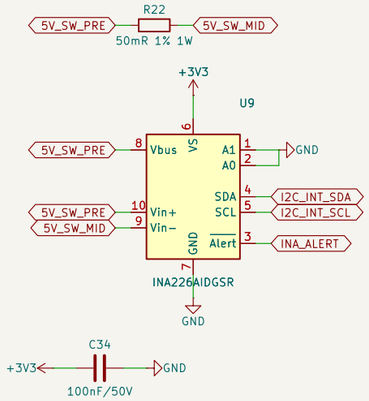
OOTY

Sheet: /mcu/
File: mcu.kicad_sch

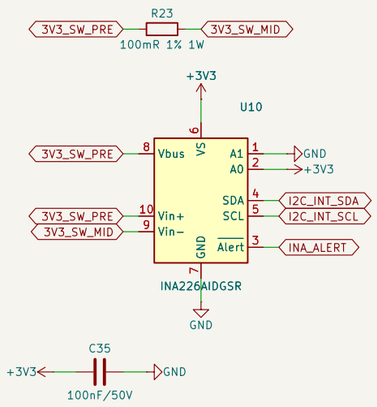
Title: MCU STM32G431MBT6

Size: A3	Date: 2026-10-20	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 3/10

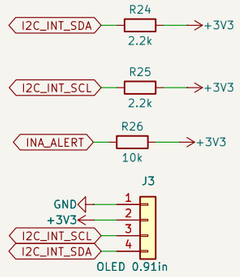
Current monitor 5V_SW



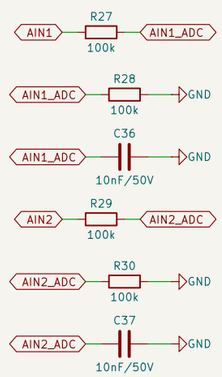
Current monitor 3V3_SW



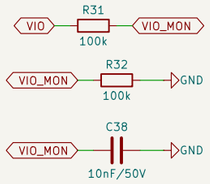
Internal I2C + OLED



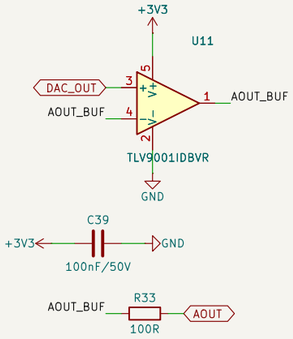
Analog inputs (0–5.5 V)



VIO monitor



DAC output buffer

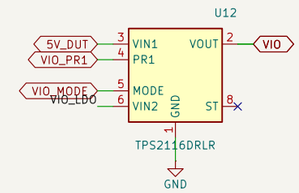


INA226 addresses: 5V_SW 0x40 (A1=A0=GND), 3V3_SW 0x41 (A0=VS). OLED 0x3C. INA228 is a drop-in replacement (same pinout).

Shunts: Kelvin connection to IN+/IN− required at layout. The low-range 100R shunt between *_MID and the rail output is on the dual-range sheet.

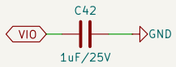
OOTY		
Sheet: /sensing/		
File: sensing.kicad_sch		
Title: Current sensing, OLED, analog I/O		
Size: A3	Date: 2026-10-20	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 4/10

VIO selection (TPS2116 mux)



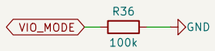




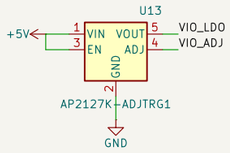






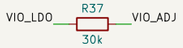


VIO LDO 1.8 / 3.3 V (AP2127K-ADJ)





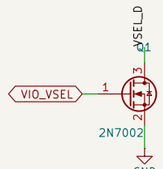












TPS2116: MODE=1 PR1=1 -> 5V_DUT ; MODE=1 PR1=0 -> LDO ; MODE=0 PR1=1 -> off (external VIO from J_PWR/J_DBG). MODE=0 PR1=0 is forbidden (selects 5 V).

LDO: Vout = 0.8 x (1 + 30k/24k) = 1.80 V ; VIO_VSEL high adds 16k in parallel -> 3.30 V.

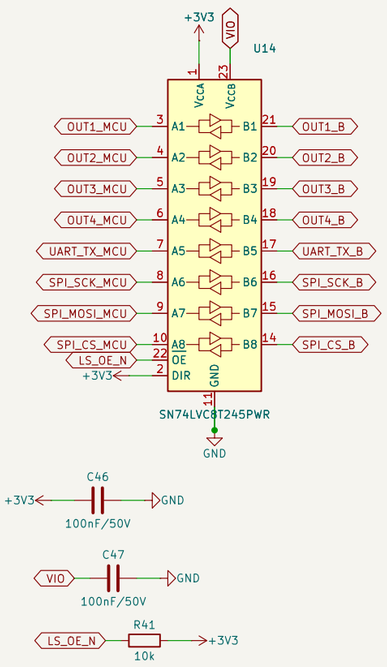
OOTY

Sheet: /vio/
File: vio.kicad_sch

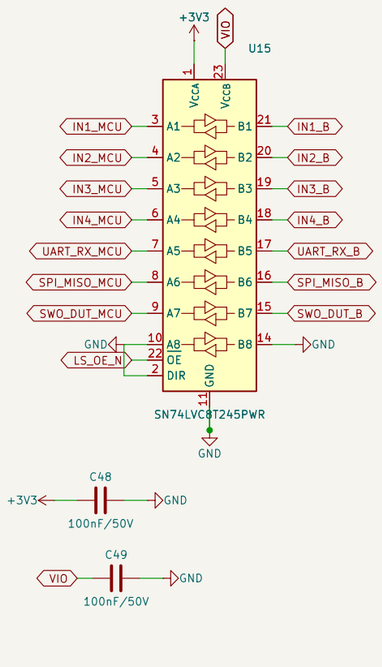
Title: VIO: 1.8 / 3.3 / 5 V or external

Size: A3	Date: 2026-10-20	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 5/10

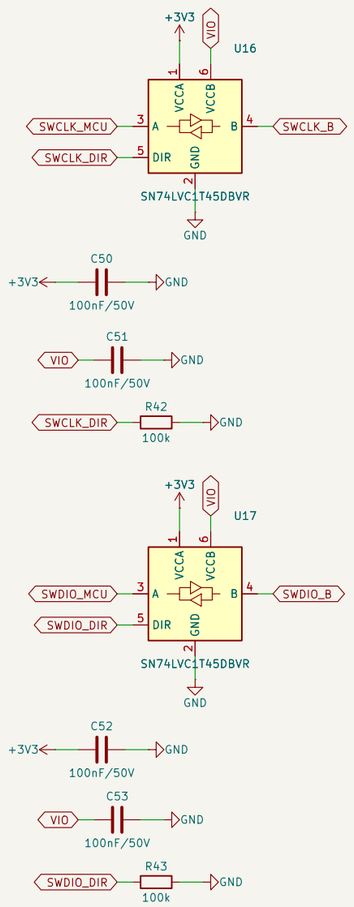
Outputs 3V3 -> VIO (DIR = A to B)



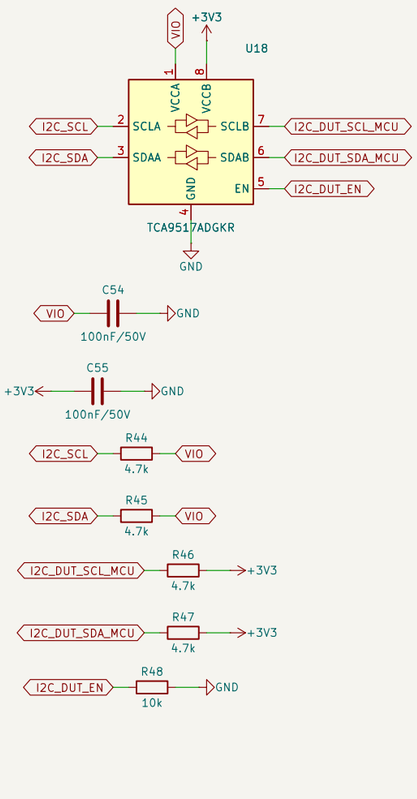
Inputs VIO -> 3V3 (DIR = B to A)



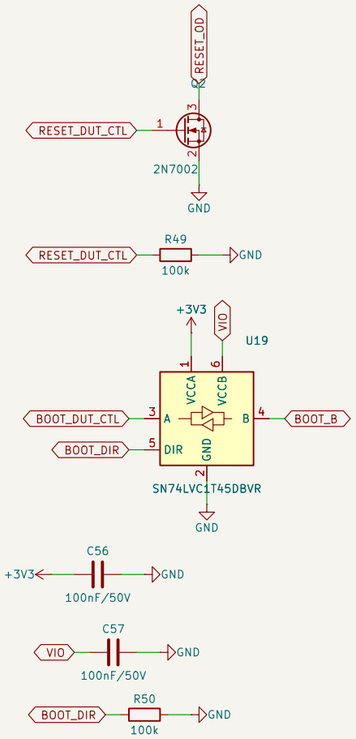
SWD probe



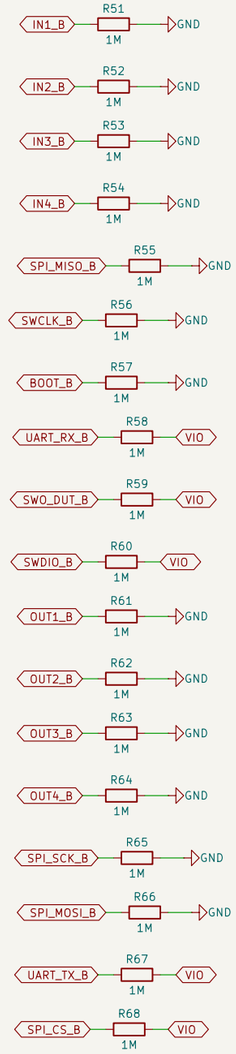
DUT I2C (A = DUT)



RESET (OD) / BOOT (3-state)



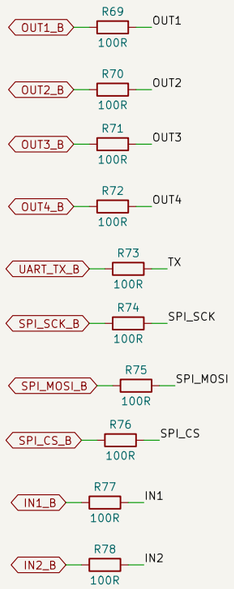
Input bias 1M



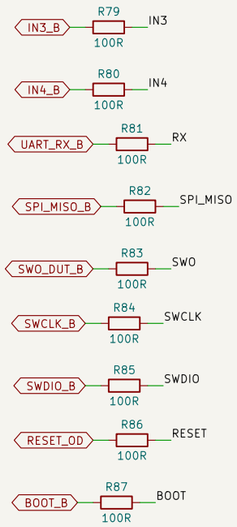
All translators are Hi-Z until firmware drives LS_OE_N low / DIR high / I2C_DUT_EN high.

OOTY		
Sheet: /translators/		
File: translators.kicad_sch		
Title: Level translators 3V3 <-> VIO		
Size: A3	Date: 2026-10-20	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 6/10

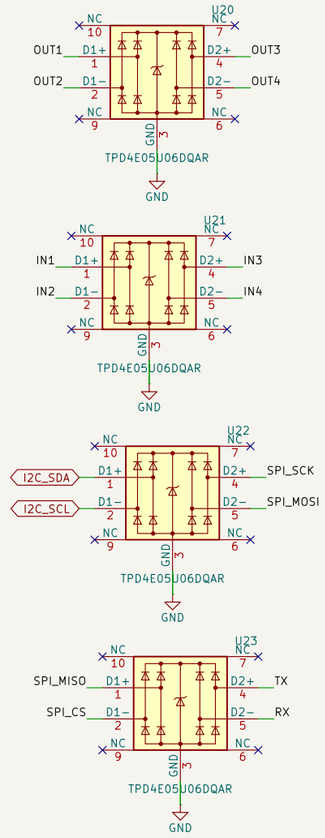
Series resistors



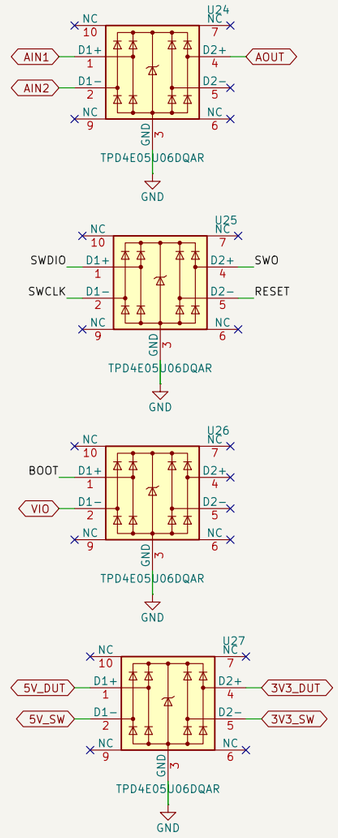
Series resistors (cont.)



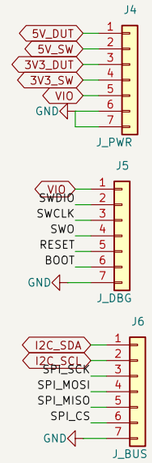
ESD (TPD4E05U06)



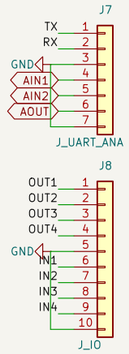
ESD (cont.)



DUT headers (2.54 mm)



DUT headers (cont.)



Mounting holes

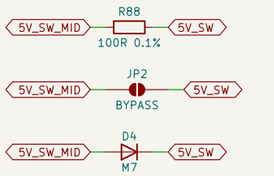


Signal path: header -> TVS -> 100R series -> translator / circuit.

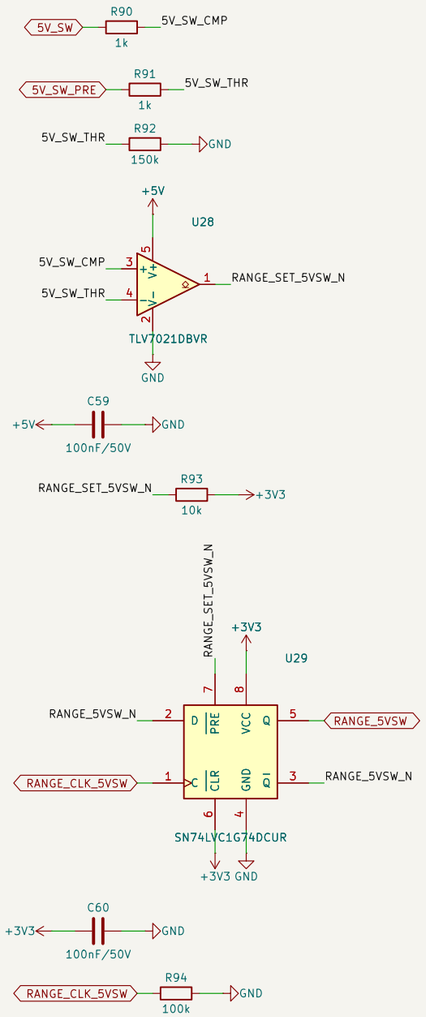
DUT rails and VIO are probed on J_PWR / J_DBG pins (no separate test points).

OOTY		
Sheet: /connectors/		
File: connectors.kicad_sch		
Title: DUT connectors and ESD		
Size: A3	Date: 2026-10-20	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 7/10

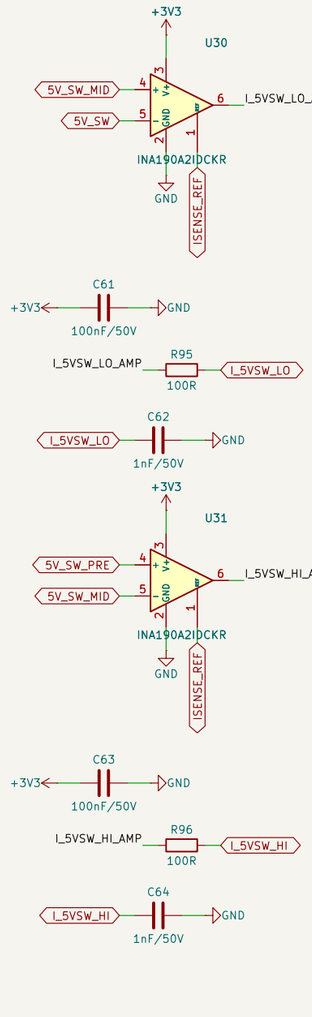
5V_SW low-range shunt and bypass



5V_SW range comparator and flip-flop



5V_SW amplifiers (x50)



5V_SW test points



LO range: 100R x 50 = 5 mV/uA. HI range: existing shunt (also read by the INA226).

Comparator trips when the drop across 100R exceeds the threshold -> /PRE low -> Q high -> 2N7002 on -> AO3401A bypasses 100R. Firmware toggles the range with CLK and reads Q.

100R, JP (BYPASS), M7 and AO3401A are in parallel between *_MID and the rail output.

M7: clamps the drop while the range switches. JP open: bridge it to bypass the 100R.

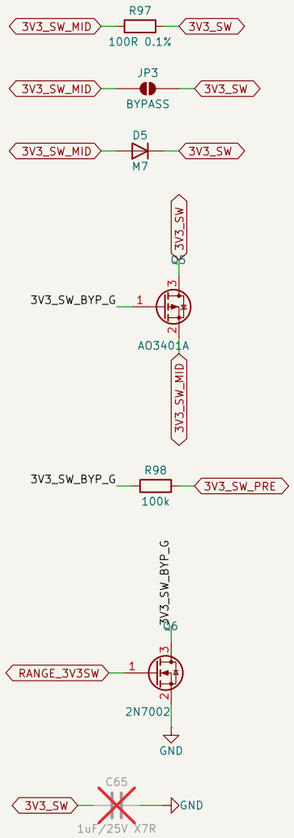
OOTY

Sheet: /range_5v_sw/
File: range_5v_sw.kicad_sch

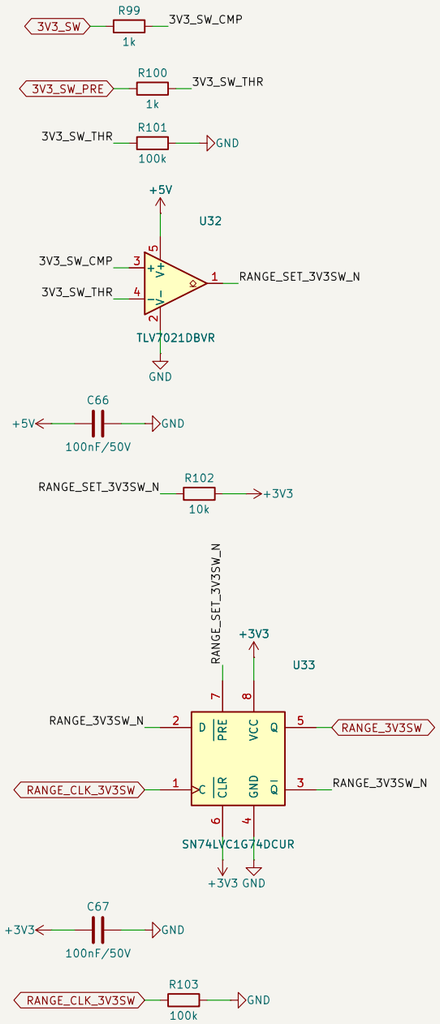
Title: Dual-range current sensing 5V_SW

Size: A3	Date: 2026-10-20	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 8/10

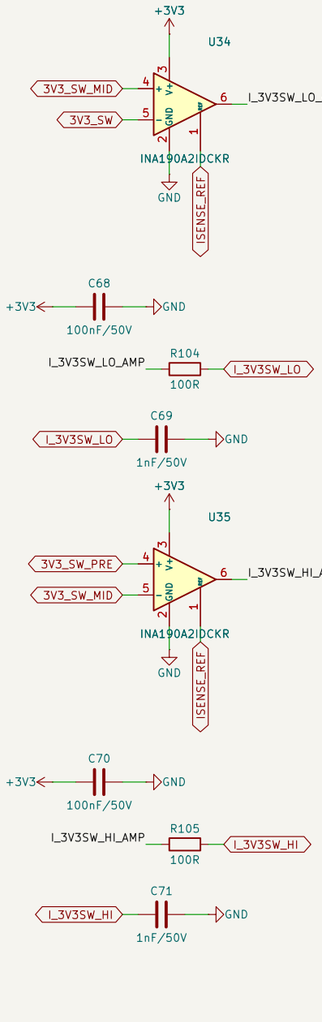
3V3_SW low-range shunt and bypass



3V3_SW range comparator and flip-flop



3V3_SW amplifiers (x50)



3V3_SW test points



LO range: 100R x 50 = 5 mV/uA. HI range: existing shunt (also read by the INA226).

Comparator trips when the drop across 100R exceeds the threshold -> /PRE low -> Q high -> 2N7002 on -> AO3401A bypasses 100R. Firmware toggles the range with CLK and reads Q.

100R, JP (BYPASS), M7 and AO3401A are in parallel between *_MID and the rail output.

M7: clamps the drop while the range switches. JP open: bridge it to bypass the 100R.

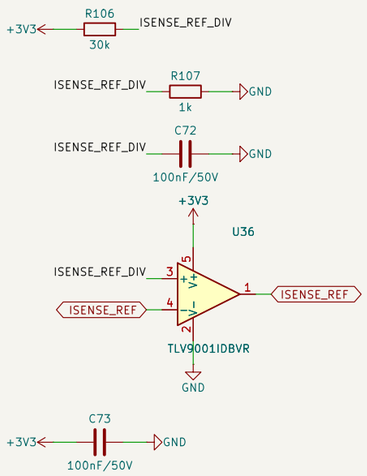
OOTY

Sheet: /range_3v3_sw/
File: range_3v3_sw.kicad_sch

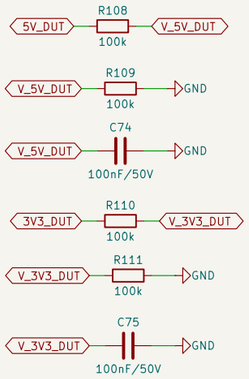
Title: Dual-range current sensing 3V3_SW

Size: A3	Date: 2026-10-20	**Rev: A**
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INA190 reference (~0.1 V)



Fixed-rail voltage readback (x0.5)



ISENSE_REF = 3.3 V x 1k / 31k = 106 mV, buffered: zero-current output of the INA190 stays in its linear range (SBOS863D 7.4).

V_5V_DUT / V_3V3_DUT: rail / 2, slow monitoring (5 ms), sample at <= 1 kS/s.

OOTY

Sheet: /readback/
File: readback.kicad_sch

Title: INA190 reference, fixed-rail voltage readback

Size: A3	Date: 2026-10-20	**Rev: A**
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